

Day 14: Nets & Surface Area

LESSON + PRACTICE

MathCraft · Grade 6 Mathematics · Professor Xavier Honablué, M.Ed.

Standard: 6.G.A.4 — represent three-dimensional figures using nets made up of rectangles and triangles, and use the nets to find surface area; MP.4, MP.7

I can: unfold a prism into a net, name every face, find the area of each face, and add them to get the surface area — and I can run it backwards to find a missing edge.

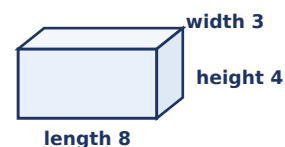
Homework: Earn 90% on IXL "Surface area of cubes and rectangular prisms" (HH, 6th grade) — [ixl.com/math/grade-6/surface-area-of-cubes-and-rectangular-prisms](https://www.ixl.com/math/grade-6/surface-area-of-cubes-and-rectangular-prisms)

Remember: show ALL work by hand in your math notebook (graph paper is best) and label every answer with units.

Learn

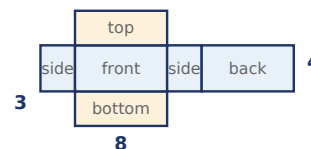
1 · A net is a 3-D figure unfolded flat

Cut a cereal box along its edges and flatten it: what you get is a **net**. Every face of the box becomes a flat shape you can measure with the area formulas you already know. **Surface area** is the total area of all the faces — the amount of cardboard, wrapping paper, glass, or paint the outside of the figure needs. It is measured in **square units**, never cubic units.



2 · A rectangular prism has 6 faces in 3 matching pairs

Top matches bottom ($l \times w$), front matches back ($l \times h$), and the two ends match ($w \times h$). So instead of six separate multiplications: **SA** = $2(l \times w) + 2(l \times h) + 2(w \times h)$. Finding all six faces on the net first and then adding is always allowed — the formula is just a shortcut for that same list.

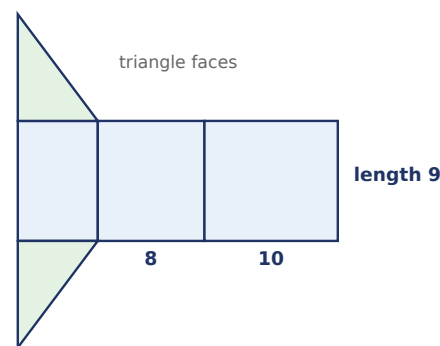


3 · A cube is the easy case

Every face of a cube is the same square, and there are 6 of them: **SA** = $6 \times s \times s$. A cube with edge 5 cm has $6 \times 25 = 150$ sq cm of surface.

4 · A triangular prism: 2 triangles + 3 rectangles

Unfold a triangular prism and you get the two triangular ends plus three rectangles — one for each side of the triangle. Each rectangle is (that side of the triangle) $\times$ (the length of the prism). So **SA** = $2 \times (\frac{1}{2}bh) + (\text{perimeter of the triangle}) \times (\text{length})$.



Worked Examples

Example 1. Find the surface area of a cube with edge length 5 cm.

- Step 1: Each face is $5 \times 5 = 25$ sq cm.
- Step 2: A cube has 6 identical faces.
- Step 3: $6 \times 25 = 150$.

✓ **Surface area = 150 sq cm**

Example 2. Find the surface area of a rectangular prism 8 in long, 3 in wide, and 4 in tall.

- Step 1: Top and bottom: $8 \times 3 = 24$ each $\rightarrow 2 \times 24 = 48$.
- Step 2: Front and back: $8 \times 4 = 32$ each $\rightarrow 2 \times 32 = 64$.
- Step 3: Two ends: $3 \times 4 = 12$ each $\rightarrow 2 \times 12 = 24$.
- Step 4: Add all six faces: $48 + 64 + 24 = 136$.

✓ **Surface area = 136 sq in**

Example 3. A triangular prism is 9 ft long. Its triangular end is a right triangle with legs 6 ft and 8 ft and a hypotenuse of 10 ft. Find the surface area.

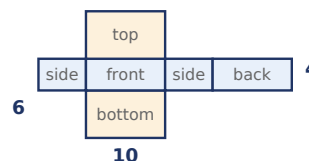
- Step 1: Two triangular ends: each is $\frac{1}{2} \times 6 \times 8 = 24$, so $2 \times 24 = 48$.
- Step 2: The three rectangles use the triangle's three sides: $6 + 8 + 10 = 24$ ft around.
- Step 3: Rectangles together: $24 \times 9 = 216$.
- Step 4: Total: $48 + 216 = 264$.

✓ **Surface area = 264 sq ft**

Example 4. A gift box is 10 in by 6 in by 4 in. How much wrapping paper covers it exactly (no overlap)?

- Step 1: $SA = 2(10 \times 6) + 2(10 \times 4) + 2(6 \times 4)$.
- Step 2: $2(60) = 120$; $2(40) = 80$; $2(24) = 48$.
- Step 3: $120 + 80 + 48 = 248$.

✓ **248 sq in of paper**



Practice — show your work in your notebook

1. Find the surface area of a cube with edge 7 m.

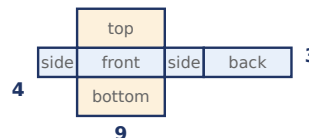
Answer: _____

2. Find the surface area of a rectangular prism 5 cm by 5 cm by 2 cm.

Answer: _____

3. The net shown folds into a rectangular prism that is 9 in long, 4 in wide, and 3 in tall. Find its surface area.

Answer: _____



4. A triangular prism is 10 cm long. Its ends are right triangles with legs 3 cm and 4 cm and hypotenuse 5 cm. Find the surface area.

Answer: _____

5. How many faces does the net of a rectangular prism have, and how do they pair up? Explain in a sentence.

Answer: _____

6. **Find the mistake.** For a prism 6 ft by 4 ft by 5 ft, Tasha wrote $SA = (6 \times 4) + (6 \times 5) + (4 \times 5) = 24 + 30 + 20 = 74$ sq ft. What went wrong?

Answer: _____

7. A rectangular prism is 5 in by 3 in by h in and has a surface area of 94 sq in. Find h .

Answer: _____

8. A cube has a surface area of 216 sq cm. How long is each edge?

Answer: _____

9. An open-top fish tank is 20 in long, 10 in wide, and 12 in tall. How much glass does it take (no lid)?

Answer: _____

10. Find the surface area of a box that is 2.5 ft by 4 ft by 3 ft.

Answer: _____

11. A storage crate is 12 ft by 8 ft by 5 ft. One can of paint covers 50 sq ft. How many whole cans are needed to paint the outside?

Answer: _____

12. Two nets are drawn: one has 6 squares in a row, the other has 4 squares in a row with 1 square attached above and 1 below the second square. Which one folds into a cube? Explain.

Answer: _____

Answer Key & Worked Solutions

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1. Find the surface area of a cube with edge 7 m.

- Step 1: One face: $7 \times 7 = 49$.
- Step 2: Six faces: $6 \times 49 = 294$.

✓ **Answer: 294 sq m**

2. Find the surface area of a rectangular prism 5 cm by 5 cm by 2 cm.

- Step 1: $2(5 \times 5) = 50$.
- Step 2: $2(5 \times 2) = 20$.
- Step 3: $2(5 \times 2) = 20$.
- Step 4: $50 + 20 + 20 = 90$.

✓ **Answer: 90 sq cm**

3. The net shown folds into a rectangular prism that is 9 in long, 4 in wide, and 3 in tall. Find its surface area.

- Step 1: $2(9 \times 4) = 72$.
- Step 2: $2(9 \times 3) = 54$.
- Step 3: $2(4 \times 3) = 24$.
- Step 4: $72 + 54 + 24 = 150$.

✓ **Answer: 150 sq in**

4. A triangular prism is 10 cm long. Its ends are right triangles with legs 3 cm and 4 cm and hypotenuse 5 cm. Find the surface area.

- Step 1: Two triangles: $2 \times (\frac{1}{2} \times 3 \times 4) = 2 \times 6 = 12$.
- Step 2: Perimeter of the triangle: $3 + 4 + 5 = 12$ cm.
- Step 3: Rectangles: $12 \times 10 = 120$.
- Step 4: $12 + 120 = 132$.

✓ **Answer: 132 sq cm**

5. How many faces does the net of a rectangular prism have, and how do they pair up? Explain in a sentence.

- Step 1: A rectangular prism has 6 faces.
- Step 2: They form 3 matching pairs: top/bottom, front/back, and the two ends.
- Step 3: That is why the formula has three doubled products.

✓ **Answer: 6 faces in 3 congruent pairs**

6. Find the mistake. For a prism 6 ft by 4 ft by 5 ft, Tasha wrote $SA = (6 \times 4) + (6 \times 5) + (4 \times 5) = 24 + 30 + 20 = 74$ sq ft. What went wrong?

- Step 1: Tasha found only three faces — one of each pair.
- Step 2: Every face has a twin on the opposite side, so each product is doubled.
- Step 3: Correct: $2(24) + 2(30) + 2(20) = 48 + 60 + 40 = 148$.

✓ **Answer: She forgot the matching faces; correct SA is 148 sq ft**

7. A rectangular prism is 5 in by 3 in by h in and has a surface area of 94 sq in. Find h .

- Step 1: Top and bottom: $2(5 \times 3) = 30$.
- Step 2: The other four faces: $2(5h) + 2(3h) = 10h + 6h = 16h$.
- Step 3: $30 + 16h = 94 \rightarrow 16h = 64 \rightarrow h = 4$.
- Step 4: Check: $30 + 2(20) + 2(12) = 30 + 40 + 24 = 94$. ✓

✓ **Answer: $h = 4$ in**

8. A cube has a surface area of 216 sq cm. How long is each edge?

- Step 1: 6 faces, so one face is $216 \div 6 = 36$ sq cm.
- Step 2: A face is a square: $6 \times 6 = 36$, so the edge is 6 cm.

✓ **Answer: 6 cm**

9. An open-top fish tank is 20 in long, 10 in wide, and 12 in tall. How much glass does it take (no lid)?

- Step 1: Bottom: $20 \times 10 = 200$.
- Step 2: Front and back: $2(20 \times 12) = 480$.
- Step 3: Two ends: $2(10 \times 12) = 240$.
- Step 4: $200 + 480 + 240 = 920$. (The missing top would have added another 200.)

✓ **Answer: 920 sq in of glass**

10. Find the surface area of a box that is 2.5 ft by 4 ft by 3 ft.

- Step 1: $2(2.5 \times 4) = 2(10) = 20$.
- Step 2: $2(2.5 \times 3) = 2(7.5) = 15$.
- Step 3: $2(4 \times 3) = 24$.
- Step 4: $20 + 15 + 24 = 59$.

✓ **Answer: 59 sq ft**

11. A storage crate is 12 ft by 8 ft by 5 ft. One can of paint covers 50 sq ft. How many whole cans are needed to paint the outside?

- Step 1: $2(12 \times 8) = 192$.
- Step 2: $2(12 \times 5) = 120$.
- Step 3: $2(8 \times 5) = 80$.
- Step 4: $SA = 192 + 120 + 80 = 392$ sq ft.
- Step 5: $392 \div 50 = 7.84$, so round up.

✓ **Answer: 8 cans (392 sq ft)**

12. Two nets are drawn: one has 6 squares in a row, the other has 4 squares in a row with 1 square attached above and 1 below the second square. Which one folds into a cube? Explain.

- Step 1: A cube net needs 6 squares, but they must be arranged so no two faces overlap when folded.
- Step 2: Six squares in a single row curls around and overlaps — it does not close into a cube.
- Step 3: The 4-in-a-row net with one square above and one below folds correctly: the row wraps around the sides, and the two extra squares become the top and bottom.

✓ **Answer: The second net (4 in a row with one above and one below)**